

MINOT INTL, ND, USA

WMO: 727676

Lat: 48.255N

Lon: 101.273W

Elev: 508

StdP: 95.37

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 24013

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.6	-25.0	-31.6	0.2	-27.3	-28.9	0.3	-24.8	14.8	-7.3	13.8	-7.1	4.8	290	0.694

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	32.8	20.7	30.9	20.3	29.0	19.2	23.1	29.1	21.8	28.0	20.6	26.9	5.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	16.9	26.7	19.8	15.4	25.5	18.4	14.1	24.0	71.3	28.9	66.1	28.1	61.8	26.9	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.3	10.0	-30.4	36.9	3.2	2.0	-32.7	38.3	-34.6	39.5	-36.4	40.6	-38.8	42.0		
(5)			-30.6	25.4	3.1	1.7	-32.8	26.6	-34.7	27.5	-36.5	28.5	-38.7	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-10.9	-9.2	-2.5	5.7	12.6	18.1	21.4	20.6	15.3	7.1	-1.5	-8.4		
	DBStd	13.00	8.40	7.83	7.78	6.08	4.86	3.76	3.20	3.60	4.69	5.37	6.74	7.70		
	HDD10.0	2808	649	539	390	151	29	1	0	0	9	121	348	572		
	HDD18.3	4852	907	772	646	378	184	48	9	18	114	349	596	830		
	CDD10.0	1263	0	0	2	23	111	243	354	329	167	32	2	0		
	CDD18.3	265	0	0	0	1	8	41	105	88	22	1	0	0		
	CDH23.3	2934	0	0	0	16	133	438	1063	967	296	23	0	0		
	CDH26.7	1040	0	0	0	3	34	125	382	392	99	5	0	0		
(14) Wind	WSAvg	4.9	5.4	5.2	5.2	5.2	5.2	4.6	4.2	4.2	4.5	5.1	5.2	5.2		
(15) Precipitation	PrecAvg	457	17	13	21	44	61	78	66	52	42	25	18	16		
	PrecMax	721	54	38	70	205	172	202	188	161	155	117	69	48		
	PrecMin	261	1	0	0	1	0	4	2	6	3	0	0	1		
	PrecStd	116	13	11	19	45	41	38	39	39	34	28	16	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	8.9	18.9	25.6	30.2	32.6	35.1	36.1	32.9	26.3	17.5	9.0		
		MCWB	3.6	4.9	9.6	13.2	17.0	21.7	22.2	21.2	18.9	14.4	9.2	4.1		
	2%	DB	5.0	5.9	14.1	21.6	26.8	30.0	32.4	33.0	29.3	21.9	13.2	6.2		
		MCWB	2.2	3.0	7.6	11.2	15.4	19.4	21.8	20.1	17.5	12.8	7.0	2.7		
	5%	DB	3.1	3.5	10.3	18.8	24.1	27.8	30.5	30.9	26.9	18.5	10.1	3.9		
		MCWB	1.0	1.5	5.7	9.7	13.7	18.5	21.3	19.8	16.5	10.7	5.1	1.2		
	10%	DB	1.0	1.4	7.3	15.9	22.0	25.9	28.7	28.6	24.0	15.5	7.4	2.0		
		MCWB	-0.6	0.0	4.1	8.5	13.0	17.4	20.0	19.0	15.4	9.6	3.9	-0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	5.2	10.8	14.3	18.8	24.0	25.3	24.1	21.1	16.0	10.0	4.7		
		MCDB	7.0	8.1	17.1	23.8	26.3	29.2	31.0	30.2	29.1	23.0	17.1	8.1		
	2%	WB	2.4	3.3	8.1	12.2	16.8	21.4	23.6	22.1	19.2	13.5	7.5	2.9		
		MCDB	4.8	5.4	13.6	20.0	23.8	27.1	29.5	29.0	26.0	19.9	12.4	5.9		
	5%	WB	1.1	1.6	5.9	10.4	15.2	19.9	22.4	20.9	17.8	11.8	5.5	1.3		
		MCDB	3.1	3.4	10.0	17.2	21.6	25.6	28.3	27.9	24.3	17.1	9.5	3.8		
	10%	WB	-0.6	0.1	4.1	8.9	13.9	18.4	21.2	19.9	16.3	10.1	3.9	-0.1		
		MCDB	1.0	1.4	7.2	15.2	20.0	24.2	27.2	26.6	22.7	15.1	7.3	1.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.0	9.6	11.9	12.4	11.4	12.4	13.5	12.6	10.7	9.1	8.3		
		MCDBR	9.8	9.9	13.9	16.8	16.3	14.5	15.0	16.6	17.2	15.9	13.4	10.2		
		MCWBR	7.9	7.6	8.8	8.7	7.7	6.8	6.8	7.8	8.4	8.3	7.7			
		MCDBR	9.6	9.2	13.3	15.5	13.5	12.7	13.4	14.7	15.0	14.8	12.9	10.3		
	5% WB	MCWBR	7.9	7.3	8.8	8.6	7.2	7.1	7.1	7.0	7.7	8.4	8.3	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.258	0.292	0.329	0.346	0.367	0.376	0.385	0.341	0.299	0.266	0.244		
	taud		2.406	2.427	2.452	2.401	2.409	2.394	2.383	2.338	2.432	2.525	2.493	2.449		
	Ebn at Noon		844	920	935	924	916	895	879	855	865	857	821	811		
	Edh at Noon		64	79	92	108	112	115	114	114	94	73	59	54		
(44) All-Sky Solar Radiation	RadAvg		1.59	2.69	3.89	4.89	5.76	6.22	6.61	5.42	4.01	2.51	1.51	1.12		
	RadStd		0.14	0.27	0.31	0.36	0.41	0.45	0.27	0.38	0.29	0.28	0.22	0.12		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.61	-0.98	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page